



Factor-Based Risk Parity for US Public Pension Funds

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Introduction

A framework for reallocating pension fund portfolios to achieve systematic risk equilibrium.

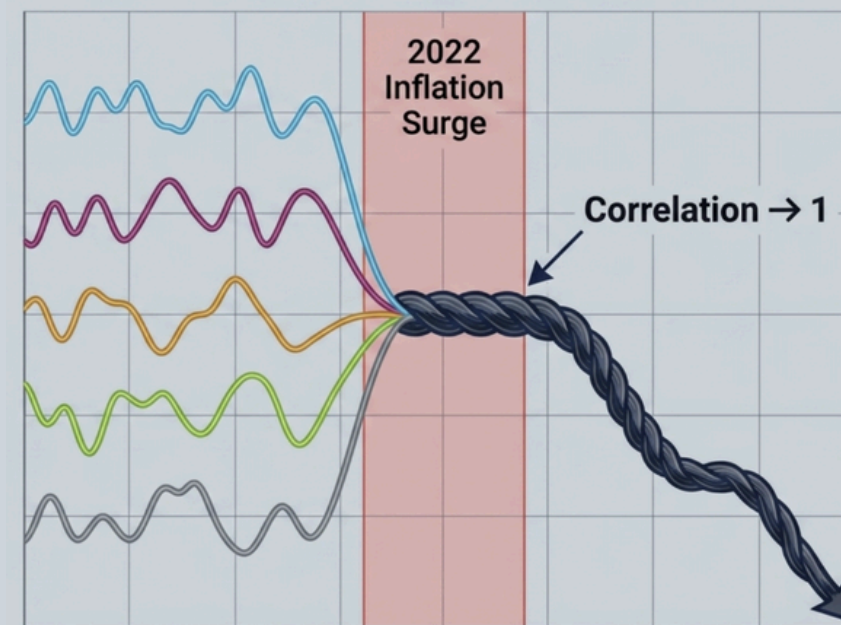
Asset-Class Based Diversification Carries a Structural Vulnerability

Asset-class diversification is often misleading. Distinct asset classes share massive overlaps in their underlying macroeconomic exposures.



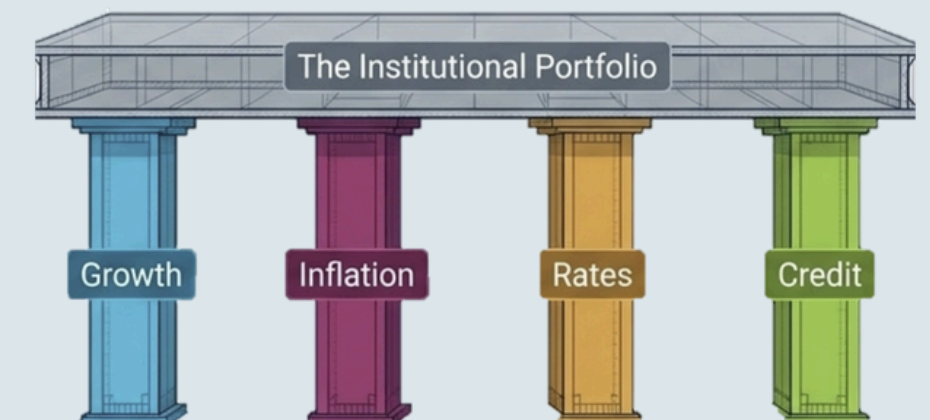
When Macro Shocks Strike, it Stops Yielding Protection

During the 2022 macro shock, cross-asset correlations rapidly converged toward 1, neutralizing the supposed benefits of holding a wide variety of traditional asset classes.



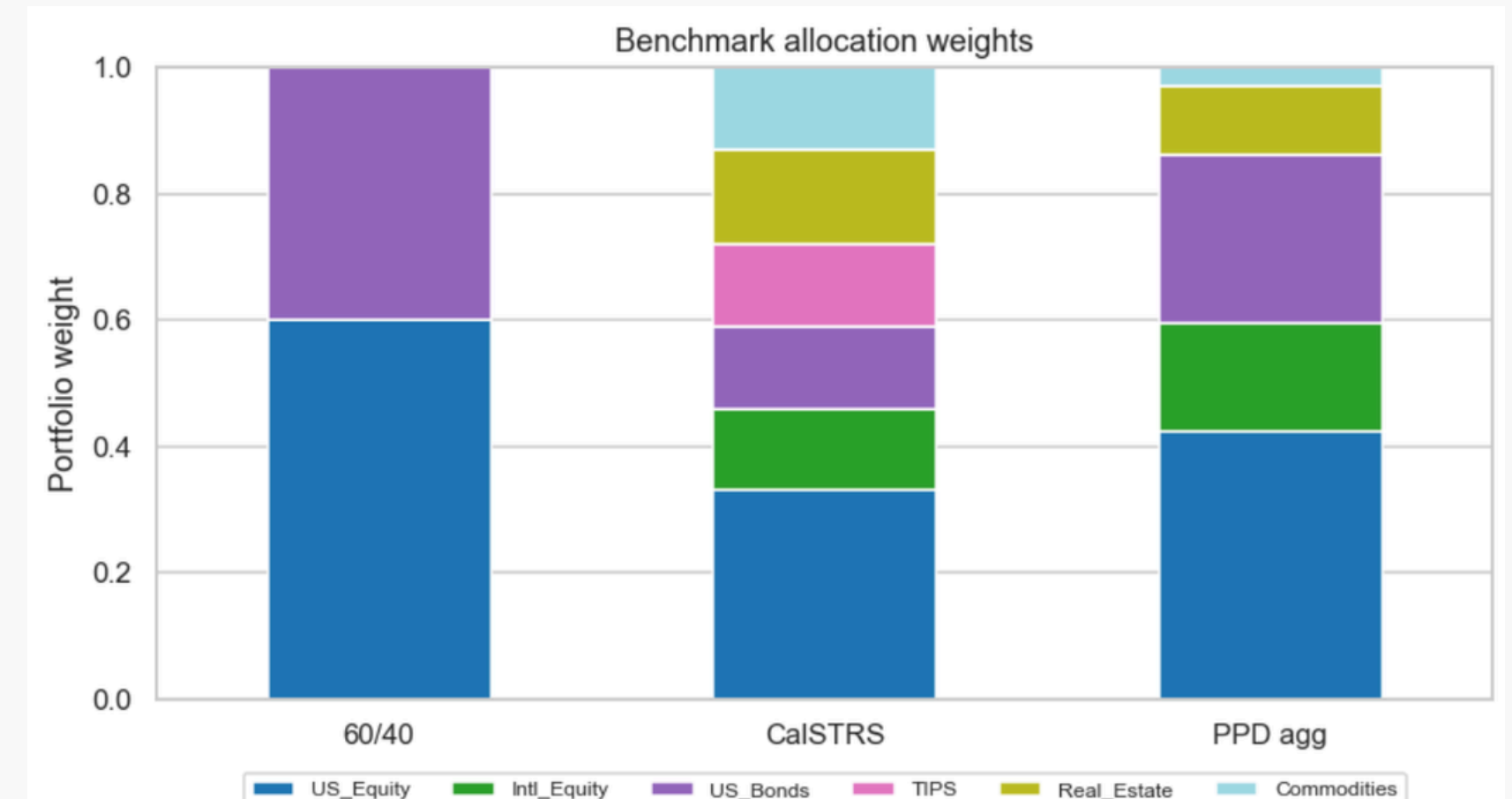
A New Foundation: Macro Factor Risk Parity

Construct a pension fund allocation such that the four key macro risk factors – Growth, Inflation, Real Rates and Credit Spreads – each contributes an equal share of the portfolio's total systematic risk.



Data and Portfolio Proxies

- Monthly panel: Jan 2004–Jan 2026
- 264 aligned observations
- Benchmarks reweighted into six proxy asset classes
- Macro factors used to estimate portfolio risk exposures



Asset-Class Proxies

- US Equity
- Intl Equity
- US Bonds
- TIPS
- Real Estate
- Commodities

Benchmark Portfolios

- 60/40 benchmark
- CalSTRS-style allocation
- PPD FY2024 aggregate
- Mapped to Asset-Class proxies

Macro Factors

- Growth: IP growth
- Inflation: CPI
- Real Rates: TIPS yield Δ
- Credit: Baa–Treasury Δ

Methodology



Estimate Factor Exposures

Regress each asset-class return on four macro factors

Obtain factor loading matrix B

Translate asset weights into macro risk exposures

$$r_{i,t} = \alpha_i + \beta_i F_t + \epsilon_{i,t}$$

Optimize FBRP Portfolio

Choose long-only weights across six asset proxies

Target balanced systematic factor-risk shares

Apply pension-style allocation bounds

$$\min_w \sum_k (s_k(w) - 1/K)^2$$



Decompose Portfolio Risk

Split asset covariance into systematic and residual risk

Measure each factor's contribution to portfolio variance

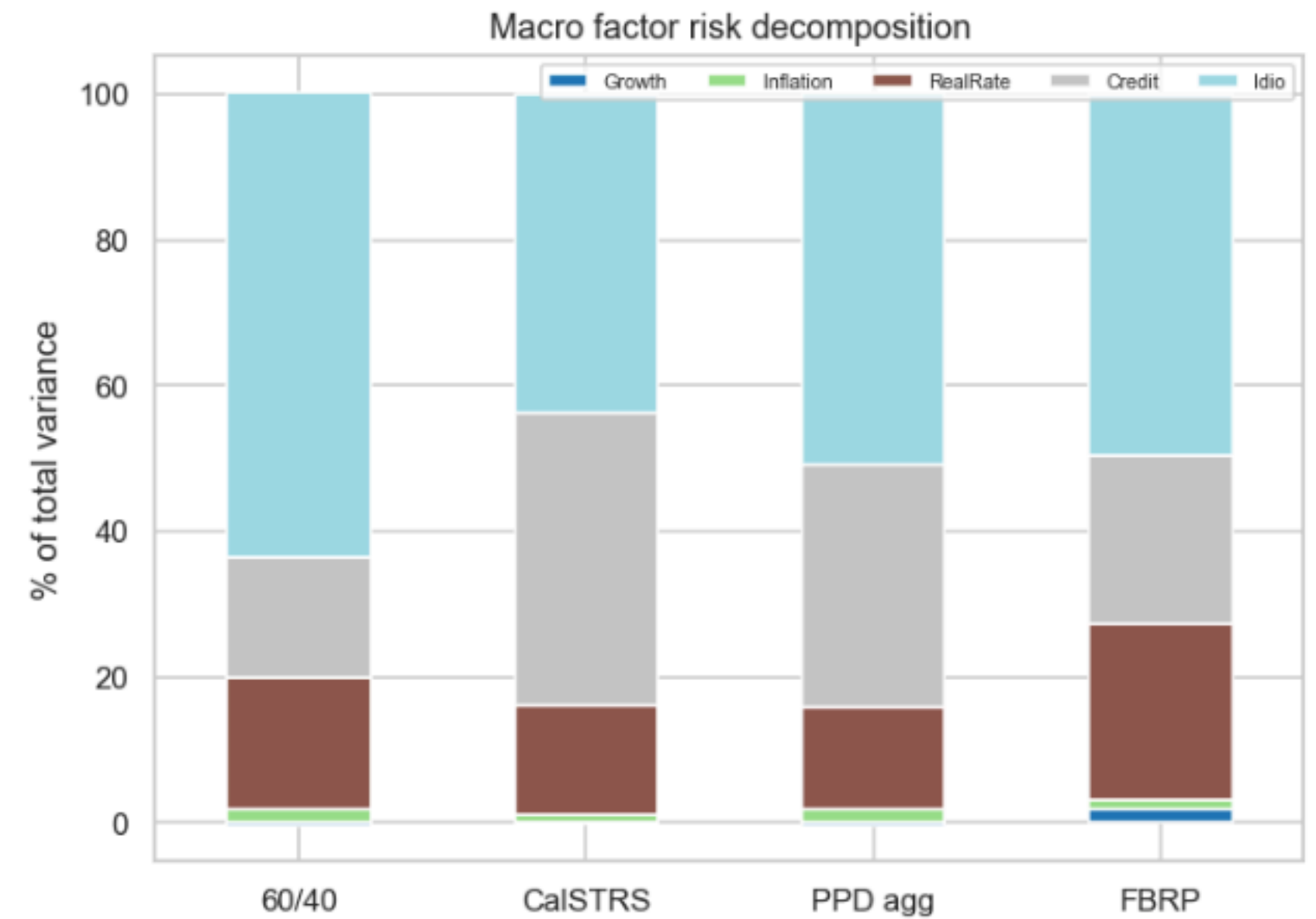
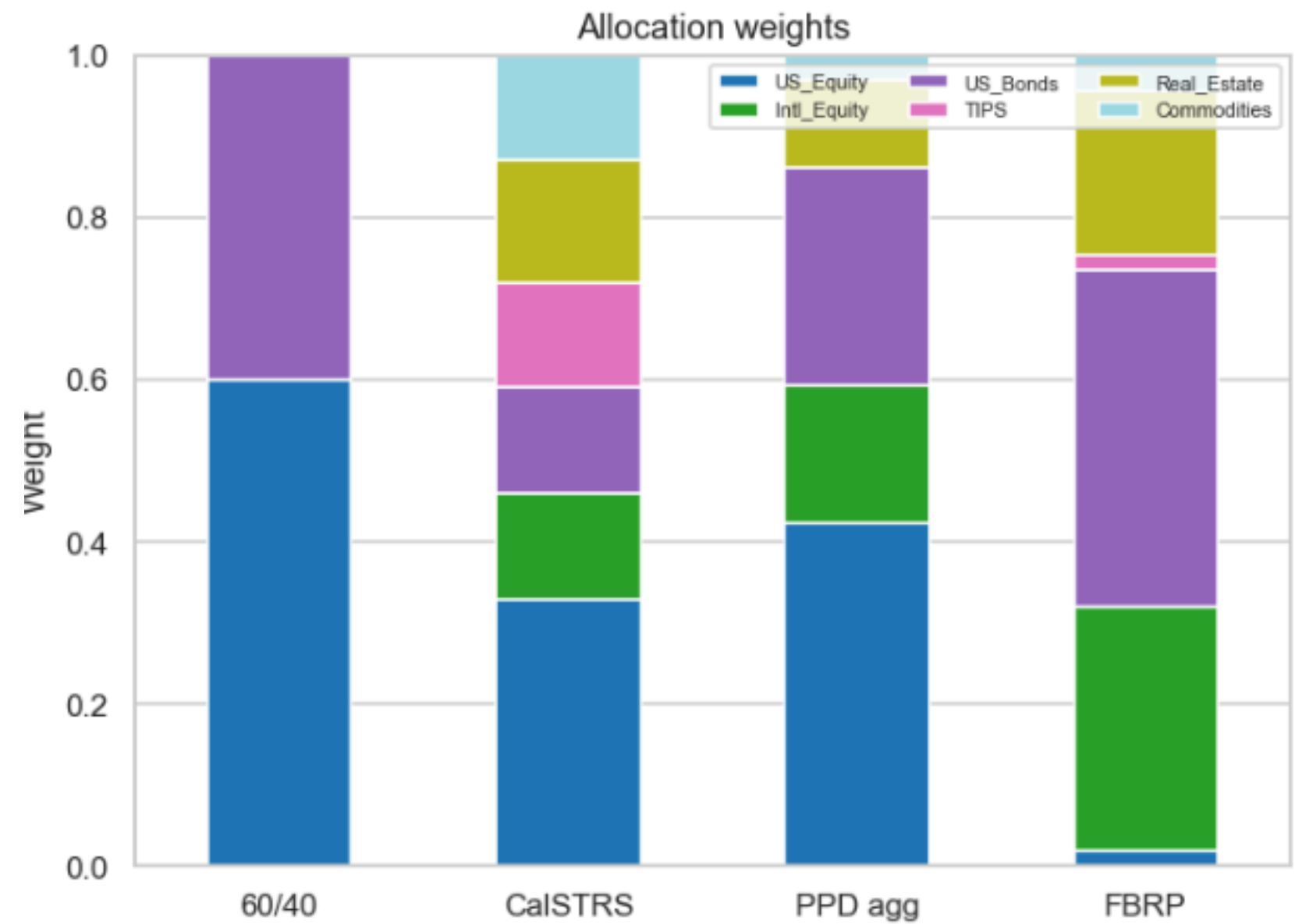
Diagnose hidden concentration in traditional allocations

$$\Sigma_a = B \Sigma_f B^\top + D$$

Key idea: Diversification is evaluated by macro factor risk contributions, not by asset-class labels.

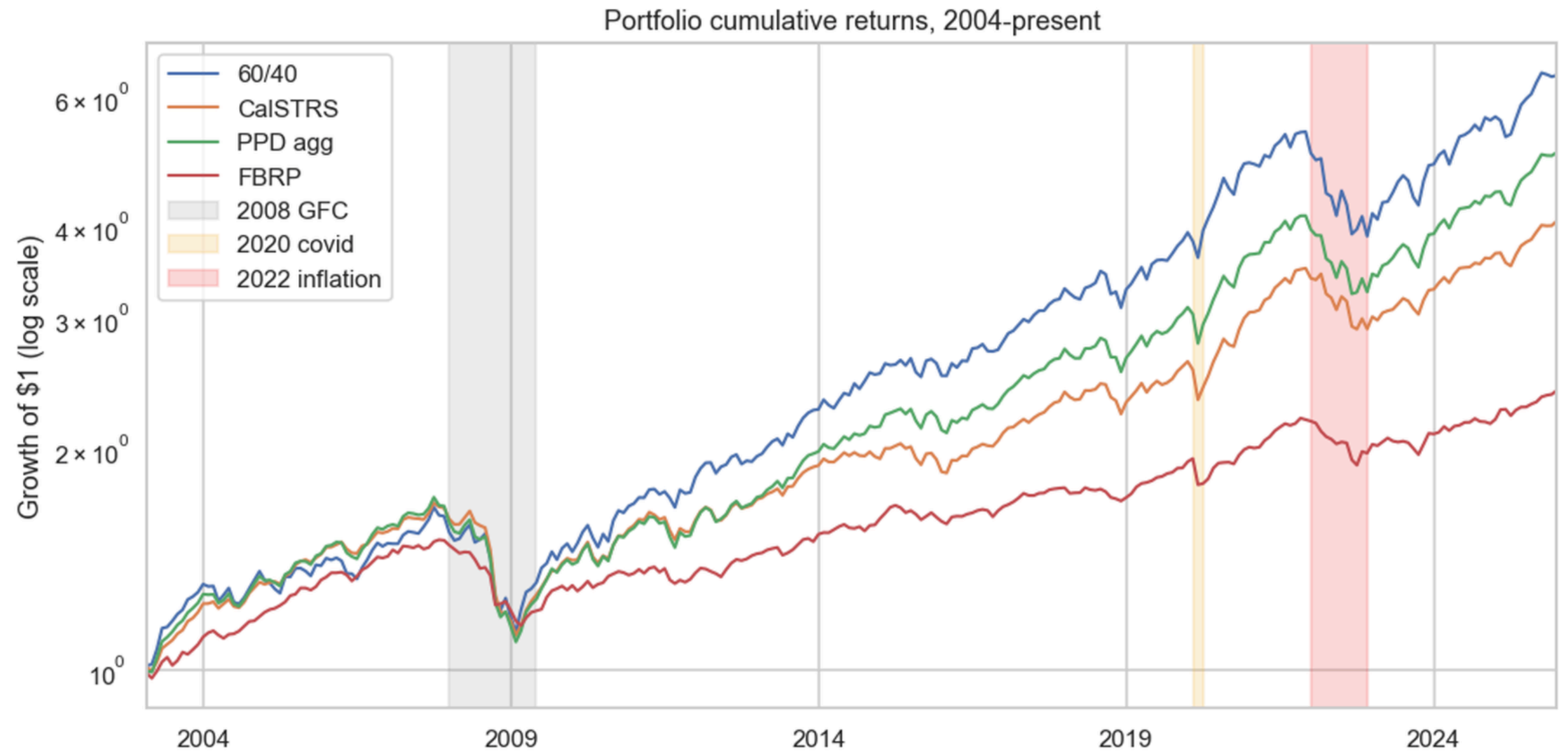


Proposed Asset Allocation

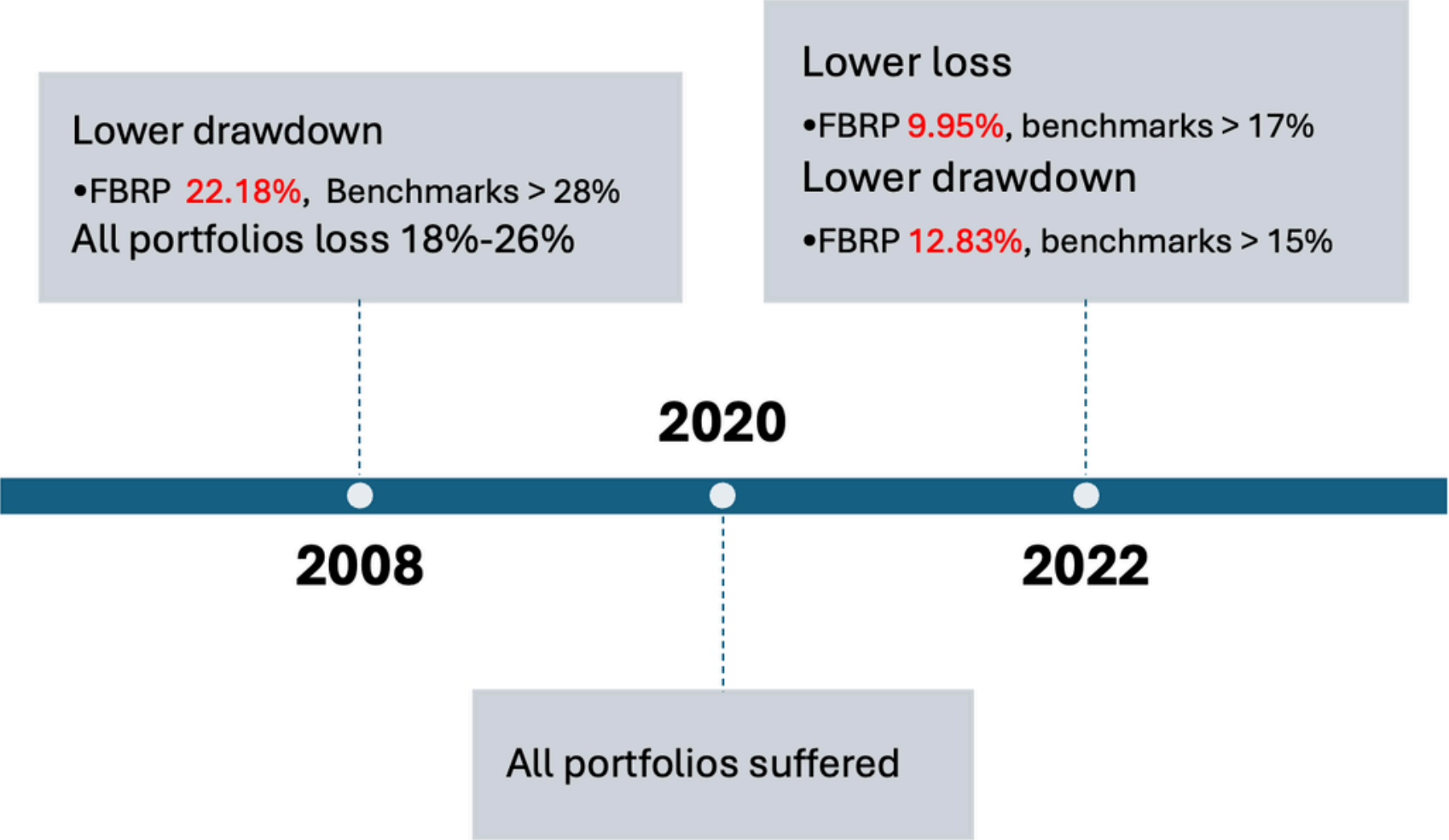


Backtesting from 2004 to 2026

- Lower volatility
 - FBRP **5.3%**
 - Benchmarks 8.0%~10.6%
- Lower drawdown
 - FBRP **23.5%**
 - Benchmarks 32% ~36%



Stress-period Performance



Conclusion

Traditional Portfolios:

- Diversified across assets
- Hidden Macro Risk Concentration

Factor-Based Risk Parity:

- Balanced Macro Factor Exposure
- Lower Drawdowns & Tail Risk
- Stronger Downside Resilience
- Improved Pension Stability

Caveats

- In-Sample Optimization
- Simplified Factor Proxies
- Smoothed Alternative Assets Returns



Thank you

